

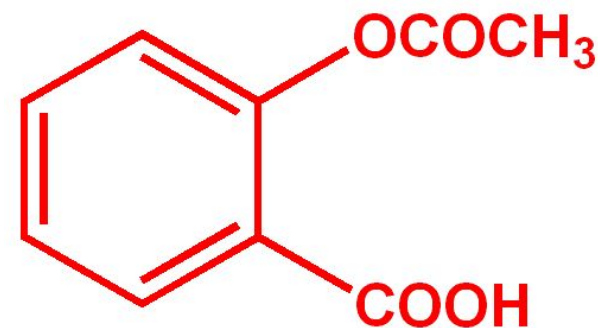
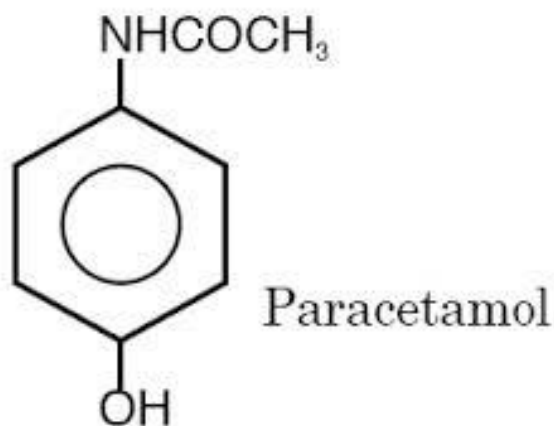
Analgesics

Analgesics reduce or abolish pain without causing impairment of consciousness, mental confusion, incoordination or paralysis or some other disturbances of nervous system.

- (i) Non-narcotic (non-addictive) analgesics
- (ii) Narcotic drugs

(i) Non-narcotic (non-addictive) analgesics

Aspirin and paracetamol belong to the class of non-narcotic analgesics.



Aspirin (pain-killer)

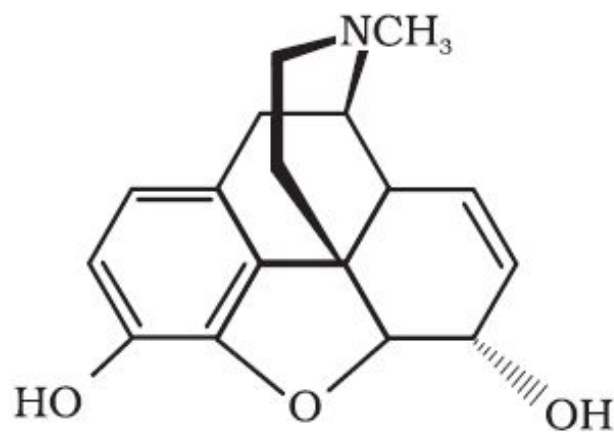
How does aspirin relieve pain?

Aspirin inhibits the synthesis of chemicals known as **prostaglandins** which stimulate inflammation in the tissue and **cause pain**. These drugs are effective in relieving skeletal pain such as that due to arthritis.

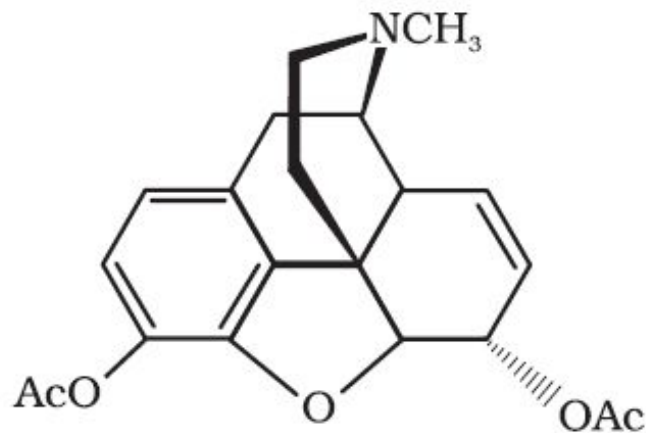
These drugs have many other effects such as reducing fever (antipyretic)

Narcotic analgesics

Morphine and many of its homologues, when administered in medicinal doses, relieve pain and produce sleep. Morphine narcotics are sometimes referred to as opiates, since they are obtained from the opium poppy.

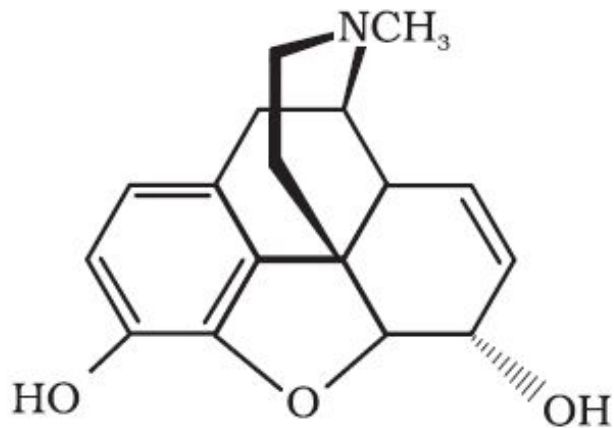


Morphine

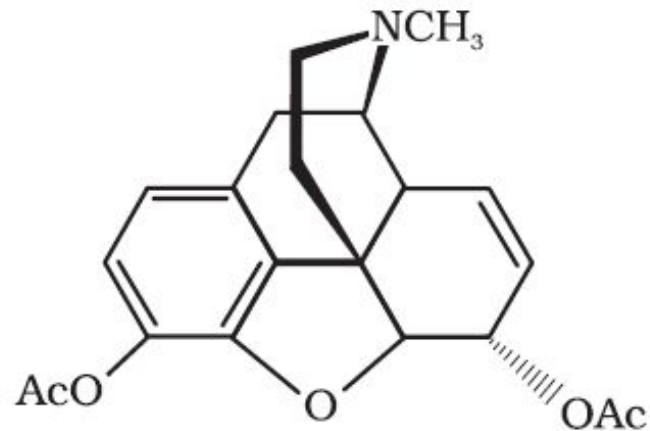


Heroin

Morphine and heroin are used for the relief of postoperative pain, cardiac pain and pains of terminal cancer, and in child birth.



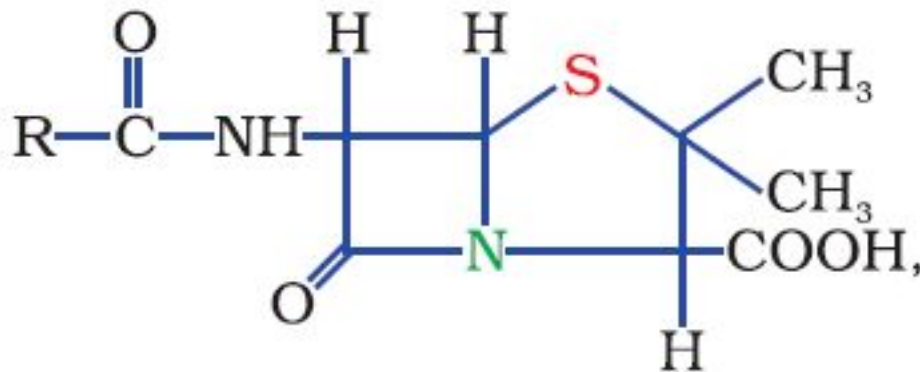
Morphine



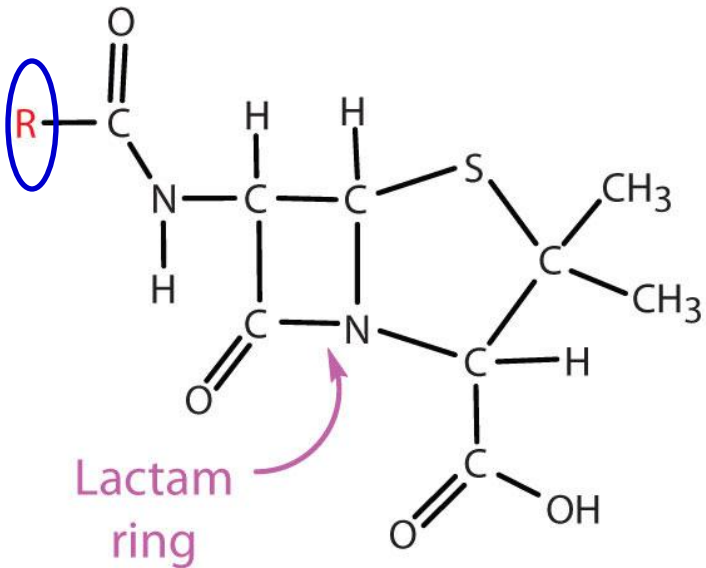
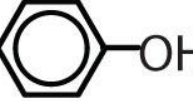
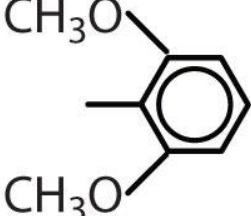
Heroin

Antibiotics

Antibiotics are used as drugs to treat infections because of their low toxicity for humans and animals.



General Structure of Penicillin

Penicillin Structure	R Group	Drug Name
 The diagram shows the general structure of a penicillin molecule. It features a central four-membered beta-lactam ring fused to a five-membered thiazolidine ring. The beta-lactam ring consists of a carbonyl group (C=O) and a nitrogen atom (N-H). The thiazolidine ring contains a sulfur atom (S) and a quaternary carbon atom bonded to two methyl groups (CH₃). A side chain is attached to the carbonyl carbon of the beta-lactam ring, represented by a red 'R' in a blue oval. A pink arrow points to the beta-lactam ring with the label 'Lactam ring'.	$\text{—CH}_2\text{—}$ 	penicillin G
	$\text{CH}_2\text{—O—}$ 	penicillin V
	—CH—  NH_2	ampicillin
	—CH—  NH_2	amoxicillin
		methicillin

Initially antibiotics were classified as chemical substances produced by microorganisms (bacteria, fungi and molds) that inhibit the growth or even destroy microorganisms

The development of synthetic methods has helped in synthesising some of the compounds that were originally discovered as products of microorganisms

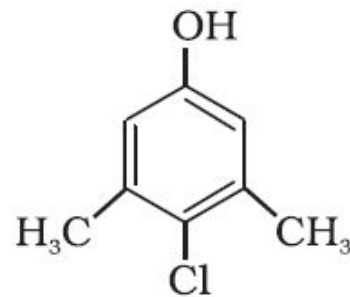
Definition of antibiotic has been modified.

An antibiotic now refers to a substance produced wholly or partly by chemical synthesis, which in low concentrations inhibits the growth or destroys microorganisms by intervening in their metabolic processes.

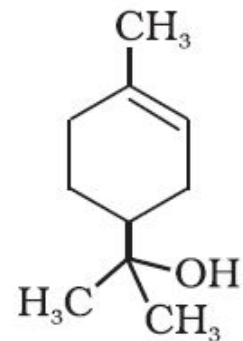
Antiseptics and disinfectants

- Antiseptics and disinfectants are also the chemicals which either **kill or prevent the growth of microorganisms**. These are not ingested like antibiotics.
- Antiseptics are applied to the living tissues** such as wounds, cuts, and diseased skin surfaces.

Commonly used antiseptic, **dettol** is a mixture of chloroxylenol and terpineol



Chloroxylenol



Terpineol

Disinfectants

Disinfectants are applied to inanimate (not alive) objects such as floors, drainage system, instruments, etc.

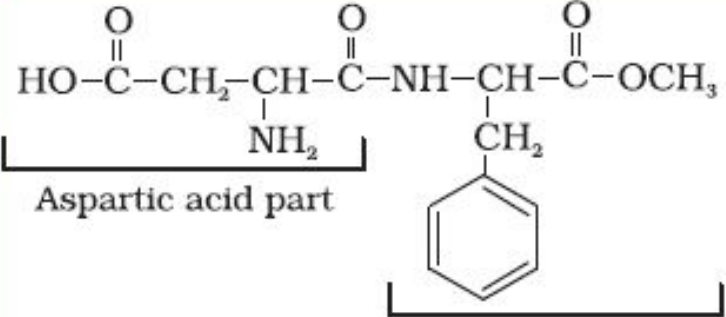
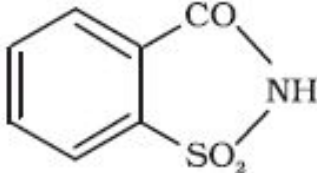
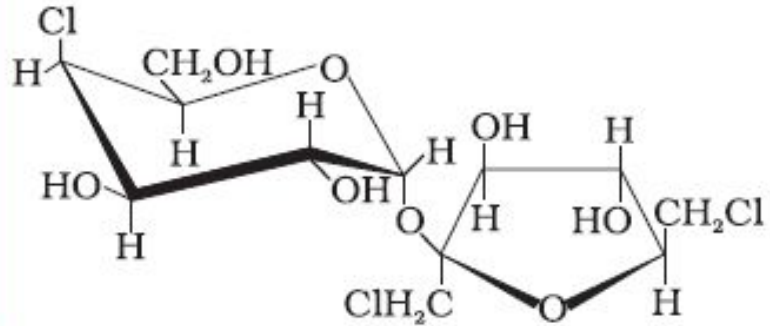
Chlorine in the concentration of 0.2 to 0.4 ppm in aqueous solution acts as disinfectant

Artificial Sweetening Agents

Natural sweeteners, e.g., sucrose add to calorie intake and therefore many people prefer to use artificial sweeteners.

Ortho-sulphobenzimide, also called saccharin, is the first popular artificial sweetening agent.

Table 16.1: Artificial Sweeteners

Artificial sweetener	Structural formula	Sweetness value in comparison to cane sugar
Aspartame	$\text{HO}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\underset{\text{NH}_2}{\text{CH}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}-\underset{\text{CH}_2}{\text{CH}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OCH}_3$  Aspartic acid part Phenylalanine methyl ester part	100
Saccharin		550
Sucralose		600

Aspartame is the most successful and widely used Artificial sweetener. It is roughly 100 times as sweet as cane sugar. It is methyl ester of dipeptide formed from aspartic acid and phenylalanine.

Use of aspartame is limited to cold foods and soft drinks because it is unstable at cooking temperature.

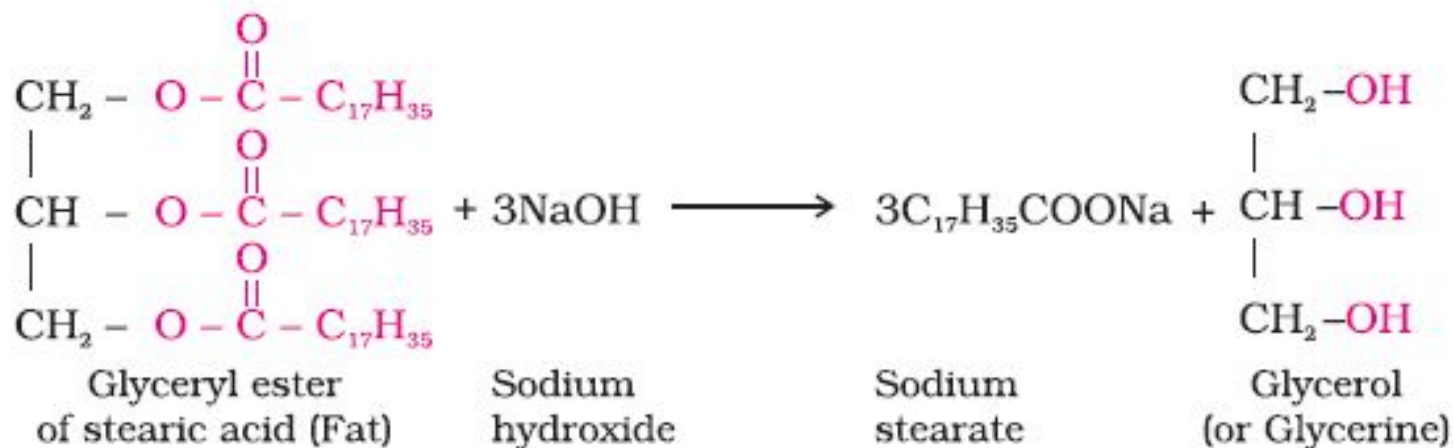
Sucralose is trichloro derivative of sucrose.

Its appearance and taste are like sugar. It is stable at cooking temperature. It does not provide calories.

Soap

Soaps are sodium or potassium salts of long chain fatty acids, e.g., stearic, oleic and palmitic acids.

Soaps containing sodium salts are formed by heating fat (*i.e., glyceryl ester of fatty acid*) with aqueous sodium hydroxide solution. This reaction is known as saponification



Toilet soaps are prepared by using better grades of fats and oils and care is taken to remove excess alkali. Colour and perfumes are added to make these more attractive.

In medicated soaps, substances of medicinal value are added. In some soaps, deodorants are added.

Shaving soaps contain glycerol to prevent rapid drying.

Detergent

Detergents are cleansing agents which have all the Properties of soaps, but which actually do not contain any soap.

These can be used both in soft and hard water as they give foam even in hard water.

Detergents are mainly classified into three categories:

- (i) Anionic detergents
- (ii) Cationic detergents and
- (iii) Non-ionic detergents

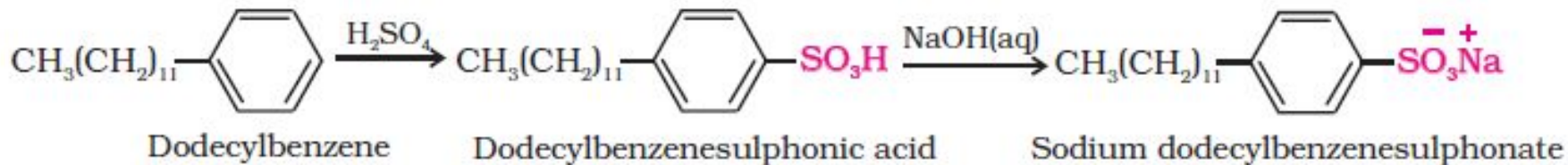
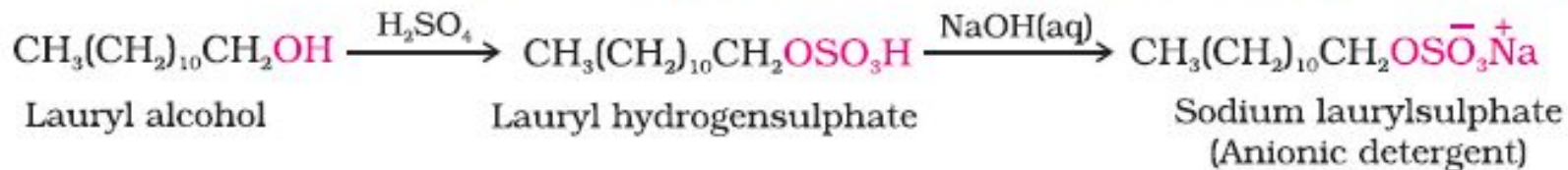
Anionic detergents

Anionic detergents are sodium salts of sulphonated long chain alcohols or hydrocarbons.

In anionic detergents, the anionic part of the molecule is involved in the cleansing action. Sodium salts of alkylbenzenesulphonates are an important class of anionic detergents.

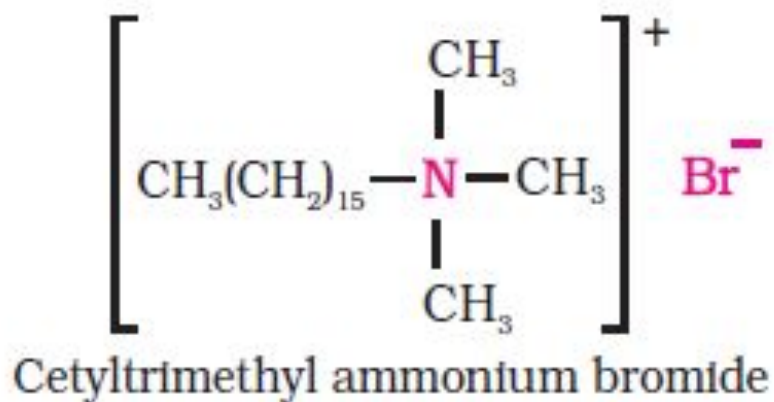
They are mostly used for household work. Anionic detergents are also used in toothpastes.

Obtained by neutralising alkyl benzene sulphonic acids with alkali.



Cationic Detergents

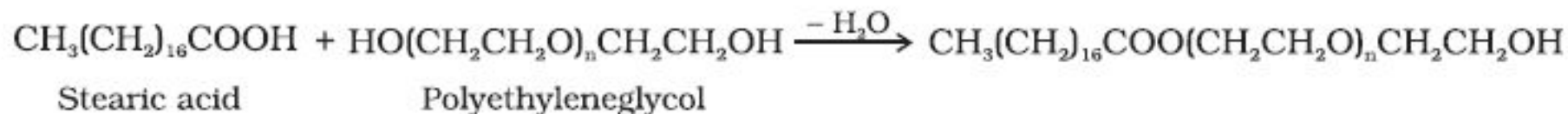
Cationic detergents are quarternary ammonium salts of amines with acetates, chlorides or bromides as anions. Cationic part possess a long hydrocarbon chain and a positive charge on nitrogen atom. Hence, these are called cationic detergents.



Cetyltrimethylammonium bromide is a popular cationic detergent and is used in hair conditioners.

Non-ionic Detergents

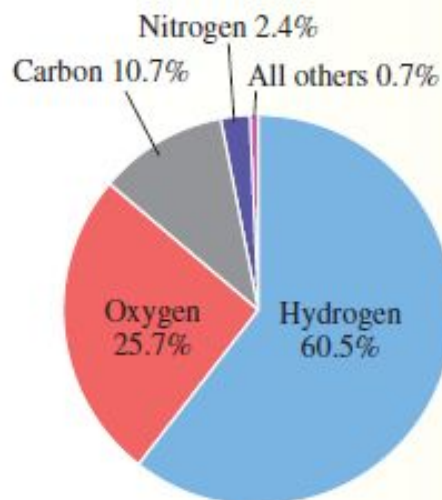
Non-ionic detergents do not contain any ion in their constitution.



Liquid dishwashing detergents are non-ionic type.

Elemental Composition of the Human Body

The distribution of elements in the human body and other living systems is very different from that found in Earth's crust. This distribution is the result of living systems *selectively* taking up matter from their external environment rather than simply accumulating matter representative of their surroundings. Food intake constitutes the primary selective intake process.



Only four elements are found in the human body at atom percent levels greater than 1%.

Hydrogen, carbon, and nitrogen are all much more abundant than in Earth's crust (Figure 1.10b), and oxygen is significantly less abundant than in Earth's crust.

The dominance of hydrogen and oxygen in the human body reflects its high water content. Hydrogen is over twice as abundant as oxygen, largely because water contains hydrogen and oxygen in a 2-to-1 atom ratio.

Carbohydrates, fats, and proteins, nutrients required by the human body in large amounts, are all sources of carbon, hydrogen, and oxygen. Proteins are the body's primary nitrogen source.

	Hydrogen	Oxygen	Carbon	Nitrogen
Water				
Carbohydrate				
Fat				
Protein				